
a
Total grey matter
Peripheral cortical grey matter
White matter
Ventricular CSF
q < 0.05
Basic covariates
CKM adjusted
BMI adjusted
Renal adjusted
Glycaemic adjusted
Vascular–lipid adjusted
Combined adjustment
0/48
−0.1
0
0.1
Standardised β
Higher module
Lower module
β range
b
Basic
CKM stage
BMI
eGFR
HbA1c + diabetes
BP + lipids
Immune system
Hemostasis
Signal transduction
Metabolism
ECM organization
Neuronal system
Higher
Lower
24
15
38
57
32
41
191
3
15
20
10
10
62
1
11
2
13
0
10
0
3
0
3
20
0
50
100
150
200
FDR-significant regional module–MRI associations
c
Protein module
Regional MRI measure
Immune system
Higher
Right thalamus
Metabolism
Lower
Left middle temporal
Right middle temporal (area)
Right middle temporal
Hemostasis
Higher
Right thalamus
Hemostasis
Lower
Right pallidum
ECM organization
Lower
Right putamen
Left pallidum
Right pallidum
Left precuneus
Left supramarginal
Right posterior cingulate
Right supramarginal
Signal transduction
Lower
Right pallidum
Neuronal system
Lower
Right pallidum
Basic covariates
CKM adjusted
BMI adjusted
Renal adjusted
Glycaemic adjusted
Vascular–lipid adjusted
Combined adjustment
−0.054 −0.046 −0.049 −0.054 −0.047 −0.047 −0.052
−0.076 −0.067 −0.066 −0.068 −0.065 −0.057 −0.061
−0.062 −0.054 −0.059 −0.060 −0.058 −0.061 −0.063
−0.065 −0.057 −0.063 −0.062 −0.061 −0.068 −0.066
−0.057 −0.050 −0.052 −0.055 −0.050 −0.050 −0.052
−0.067 −0.066 −0.063 −0.064 −0.062 −0.064 −0.064
−0.059 −0.060 −0.057 −0.059 −0.058 −0.063 −0.065
−0.071 −0.073 −0.069 −0.072 −0.069 −0.068 −0.070
−0.075 −0.075 −0.072 −0.073 −0.071 −0.071 −0.073
−0.066 −0.069 −0.066 −0.066 −0.065 −0.056 −0.057
−0.075 −0.072 −0.068 −0.071 −0.066 −0.061 −0.065
−0.063 −0.055 −0.055 −0.056 −0.056 −0.053 −0.054
−0.070 −0.066 −0.062 −0.067 −0.062 −0.055 −0.062
−0.067 −0.067 −0.063 −0.064 −0.062 −0.064 −0.064
−0.070 −0.069 −0.065 −0.065 −0.065 −0.063 −0.063
Basic covariates
CKM adjusted
BMI adjusted
Renal adjusted
Glycaemic adjusted
Vascular–lipid adjusted
Combined adjustment
q < 0.05
15/15
15/15
12/15
13/15
10/15
3/15
3/15
−0.08
0
0.08
Standardised β
q < 0.05
q ≥ 0.05
d
GFAP
NEFL
RBPMS
Basic covariates
CKM adjusted
BMI adjusted
Renal adjusted
Glycaemic adjusted
Vascular–lipid adjusted
Combined adjustment
−0.1
0
0.1
Standardised β, peripheral cortical grey-matter volume
Bootstrap distribution
β (95% CI), q < 0.05
q ≥ 0.05